POWER AND SIGNAL INTERFACE – BATTERY ENERGY STORAGE

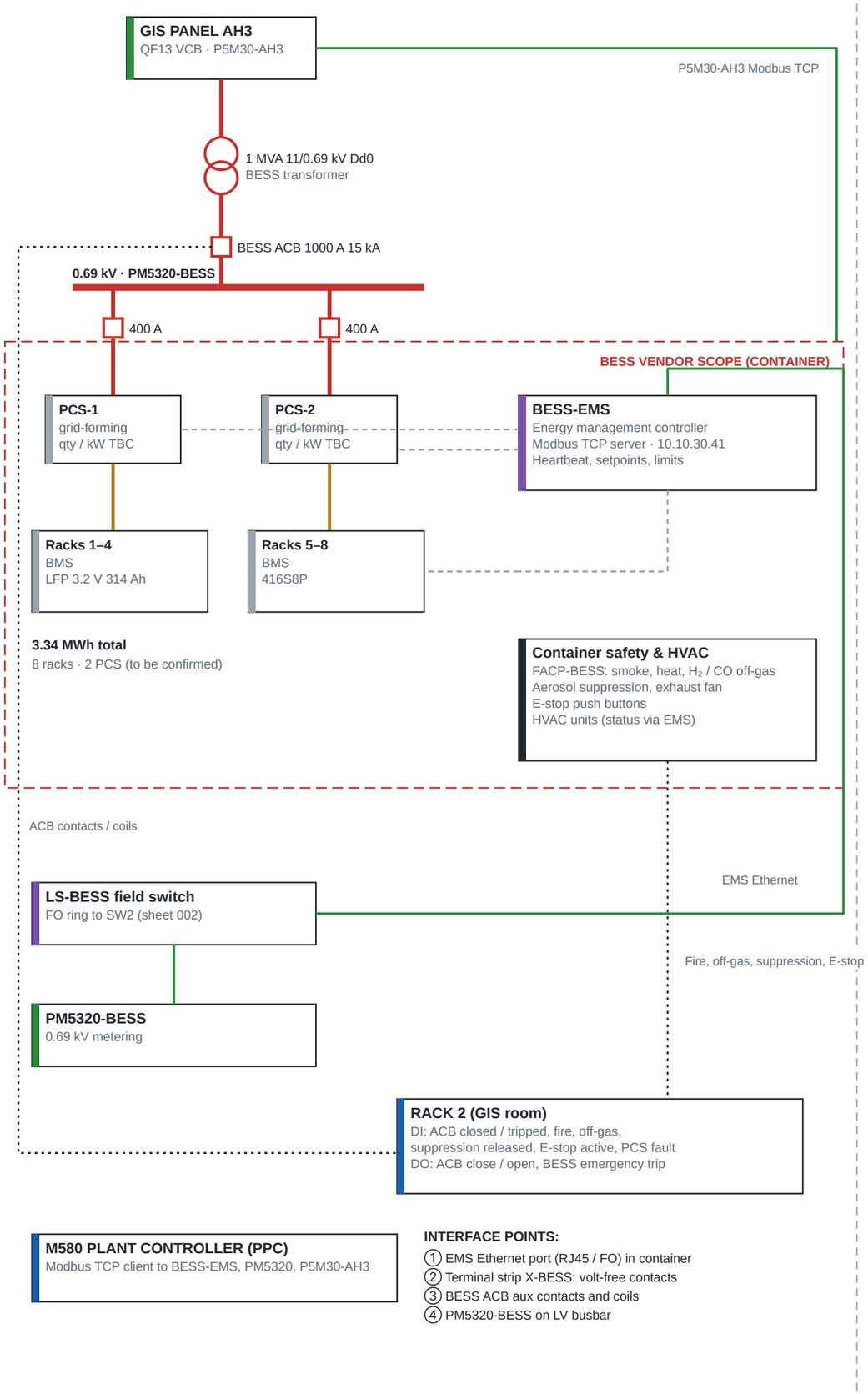


TABLE A – SIGNAL EXCHANGE SCADA ↔ BESS

Dir.	Signal	Type	Path	Rate
IN	SOC, SOH, energy stored, charged / discharged	AI comm	EMS	1 s
IN	Max charge / discharge power allowed	AI comm	EMS	200 ms
IN	P, Q, U, f per PCS; DC U, I	AI comm	EMS	200 ms
IN	Rack V, I, SOC, max temp, cell V max/min	AI comm	EMS	5 s
IN	PCS running, grid-forming, fault, warning	DI comm	EMS	200 ms
IN	HVAC temp / humidity, container status	AI comm	EMS	10 s
IN	EMS heartbeat counter	AI comm	EMS	1 s
OUT	PLC heartbeat counter	AO comm	PPC	1 s
OUT	P / Q setpoint, mode GF / GFL, start/stop	AO comm	PPC	on change
OUT	SOC limits (min / max) for PPC dispatch	AO comm	PPC	on change
IN	ACB closed / tripped	DI	Rack 2	20 ms
IN	Fire alarm, off-gas alarm, suppression released	DI	Rack 2	20 ms
IN	E-stop active, PCS common fault (backup)	DI	Rack 2	20 ms
OUT	ACB close / open, BESS emergency trip	DO	Rack 2	command
IN	LV P, Q, U, I, PF, energy	AI comm	PM5320	1 s
IN	AH3 protection, measurement, events	DI/AI comm	P5M30-AH3	1 s

Signal count (I/O list, area BESS): 6 DI · 2 DO · 88 AI comm · 16 DI comm · 8 AO comm.

TABLE B – OPERATING MODES

Mode	V / f set by	PPC role
Normal (islanded)	PCS grid-forming	Charge limit, SOC window, PV curtailment
DG support	PCS grid-forming	DG base load, BESS charges from DG
Black start	PCS grid-forming	Sequenced breaker closing (sheet 005 / HMI)
BESS unavailable	DG controller	DG isochronous, PV limited to load
Maintenance	—	BESS ACB open, AH3 earthed, permit to work

TABLE C – FAILURE BEHAVIOUR

Event	Response
EMS loses PLC heartbeat > 10 s	PCS keeps grid-forming with local droop; setpoints frozen.
PLC loses EMS heartbeat > 10 s	P1 alarm; DG auto-start; PV limit held at last value.
Fire / off-gas alarm	Hard-wired trip: PCS stop, ACB open, HVAC recirc. off,
	exhaust on. Works without communication.
E-stop in container	PCS stop and ACB trip by vendor circuit; status to SCADA.
SOC below minimum	Load shedding by LVDB priority if no DG available.
AH3 protection trip	Bus dead unless DG running; black start from BESS.

TABLE D – INTERFACE RESPONSIBILITY

Item	BESS vendor	SCADA contractor
EMS Modbus register map & heartbeat	Supply	Implement
Terminal strip X-BESS (volt-free)	Supply, wire inside	Cable to Rack 2
Fire / suppression system	Supply, commission	Monitor only
PCS control modes & droop settings	Set, document	Verify at SAT
Fibre patch panel in container	Space & power	Supply, install

LEGEND

- Power circuit
- DC circuit
- Ethernet copper – Modbus TCP / EIP
- Ethernet single-mode fibre (FO)
- Hard-wired I/O (DI / DO)
- EtherNet/IP (M580 RIO / DIO)
- Vendor-internal link (not SCADA scope)
- Vendor scope boundary

NOTES

1. Ratings from Overall SLD BFE/PR/26/01: 3.34 MWh 416S8P, 1 MVA 11/0.69 kV, 2 × 400 A to the PCS.
2. PCS quantity, rating and the EMS make are to be confirmed by the BESS vendor.
3. SCADA never releases the fire suppression; it monitors only.
4. Hard-wired trips act independently of the PLC and of communication.
5. Signal names follow the I/O list (area BESS, tags VS_BESS_…).

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A	Draft for internal review	16-OCT-2026

CLIENT:
MINISTRY OF ENERGY & WATER RESOURCES – SOMALIA
SITE: VILLA SOMALIA

CONTRACTOR:
BLUEFLAG ENERGY CO. LTD

PROJECT TITLE:
SOLARIZATION VILLA SOMALIA

SHEET TITLE:
BESS INTERFACE
Signal exchange, modes and scope boundary

SCALE: NTS	DATE: 16-OCT-2026	SIZE: A3
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DRAWN		SIGN
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